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Chahan Kropf

chahan.kropf@usys.ethz.ch

Swiss Federal Institute of Technology, ETH Zurich https://orcid.org/0000-0002-3761-2292

Lisa Vaterlaus

Swiss Federal Institute of Technology, ETH Zurich

David Bresch

Swiss Federal Institute of Technology, ETH Zurich, Switzerland https://orcid.org/0000-0002-8431-4263

Loïc Pellissier

ETH Zürich https://orcid.org/0000-0002-2289-8259

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Global vulnerability and resilience of coastal ecosystems to tropical cyclones in a warming climate

Chahan M. Kropf[1,2], Lisa Vaterlaus[1], David N. Bresch[1,2], and Loïc Pellissier[3,4]

[1]Institute for Environmental Decisions, ETH Zürich, Universitätstrasse 16, 8092 Zürich, Switzerland
[2]Federal Office of Meteorology and Climatology MeteoSwiss, Operation Center 1, P.O. Box 257, 8058 Zürich-Airport, Switzerland
[3]Institute for Terrestrial Ecosystems, ETH Zürich, Universitätstrasse 16, 8092 Zürich, Switzerland
[4]WSL, Birmensdorf Switzerland

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Abstract

Coastal ecosystems are recognized to provide a large range of services including erosion prevention, clean water provision, or protection from tropical cyclones storms. However, with climate change, the increase in frequency and intensity of tropical cyclones may alter the composition of the ecosystems themselves potentially degrading the various services they provide. While in general ecosystems are adapted to external perturbations, extreme events can promote rapid and durable shifts. In this study, we quantify the vulnerability of all ecosystems described as ecoregions at a global scale by analyzing the wind fields of tropical cyclones generated from probabilistic historical tracks. We then investigate how tropical cyclone intensity and frequency shifting in a warming climate might lead to ecosystem modifications. We show that a combined 13% of the surface of all terrestrial ecosystems is susceptible to transformation due to cyclone pattern changes between 2020 and 2050 under current policies. Even for the most resilient ecosystems already experiencing winds above 60 m/s regularly, the average interval between two storms is projected to decrease from 19 to 12 years which is potentially close to their recovery time. Our global probabilistic approach being embedded in the established IPCC climate risk frameworks enables the study of a broad temporal range, and advocates for a shift in the consideration of the tropical cyclone impact from immediate damage to effects on long-term natural recovery cycles.

1 Introduction

Species composing ecosystems have co-evolved with the geo-climatic conditions at their location. Hence, they are not only adapted to their average climatic conditions such as the temperature or precipitation regimes, but also to extreme events such as floods, wildfires, or tropical cyclones. Climate change is rapidly modifying the climatic conditions which has been shown to lead to a shift in the optimal climatic envelope

for a large number of species among insects, vertebrates, or plants [1]. This may alter larger ecosystems' compositions and lead to profound modifications of their functioning. In addition, climate change is rapidly modifying the regional and temporal patterns of extreme events [2], which might further impact global ecosystems. An increase in frequency and intensity of extreme weather events can even promote more rapid and durable shifts to ecosystems than the gradual increase of the means of climate values [3–6].

Climate change is expected to modify the intensity and frequency of tropical cyclones and their geographical impacts [7–9]. Tropical cyclones are intense low-pressure systems that form over warm ocean waters which can produce strong winds, heavy rainfall, and storm surges. As the climate warms, the surface waters of the oceans become warmer, providing more fuel for tropical storms to grow in strength. Further, climate change is expected to increase the amount of moisture in the atmosphere, which can lead to more intense rainfall during tropical storms, and the rising sea level will expose new areas to storm surges. Flooding and landslides from the heavy rainfalls, strong winds, and storm surges can cause wide-scale disturbances for the ecosystem in the path of the storm and beyond [10, 11]. This may affect all dimensions, including the ecosystem's biogeochemistry, hydrography, mobile biota, sedentary biota, or micro-biota [5, 12]. Beyond a change in cyclone severity, regional patterns are likely to change, thus exposing areas that have not historically experienced these types of storms [8, 13, 14], yielding unprecedented consequences for not only human societies [9, 15] but also for natural ecosystems [12, 16, 17].

Tropical cyclones have been shown to impact ecosystems through various processes such as wind throw, causing defoliation and canopy damage, which can lead to reduced photosynthesis, growth, and productivity of trees, as well as increased susceptibility to pest and disease outbreaks [16, 18]. Heavy rain can cause landslides, soil erosion, and nutrient loss [19] potentially causing decreased growth and productivity of vegetation, but also direct vegetation mortality [20], especially when already disturbed by anthropogenic land use [21]. Storms can add large amounts of saltwater to coastal freshwater and terrestrial ecosystems disrupting habitats [22]. Together, disturbances from tropical storms can alter the structure and composition of ecosystems, change the habitat quality for animal species, and lead to mortality or displacement of biota [23, 24], causing local loss of biodiversity [25, 26]. For instance, extreme weather events account for 11% of global mangrove forest loss between 2000 - 2020 [27]. Previous methods, such as remote sensing and field studies have sought to quantify the immediate impact of tropical storms on forests [28], but not how they may modify ecosystems over the long run. Several studies on the effect of storms on ecosystems have been conducted, most focused on explicit relation for specific species [29], part of the vegetation [17, 30], or for regional ecosystems [12]. However, estimating the potential effect of tropical cyclones on global ecosystems with a long time perspective is limited by methodological challenges [17, 31–35].

While tropical cyclones may appear as purely destructive extreme events, they also generate regular disturbances that are naturally structuring the ecosystems' dynamics [36]. The impacts of tropical cyclones are immediately followed by recovery and therefore those ecosystems are in a state of dynamic equilibrium [12, 16, 17, 31, 37]. Furthermore, tropical cyclones may support the regulation of certain ecological processes such as water provision, soil enrichment with nutrients, restoration of ecological niches, support seed dispersion, or temporarily mixing the salinity and phytoplankton content in estuaries [38]. Thus, as long as the ecosystem is resilient, tropical cyclones can also play a functional role, shaping specific ecosystems [31, 39, 40] and being a necessary perturbation to maintain their functionality, thus supporting biodiversity following the intermediate disturbance hypothesis [41–45] .

Coastal ecosystems are known to provide a large number of essential services to Human societies such as nutrient cycling, erosion control, hatchery for fishes and crustaceans, water purification, cultural importance, tourism, carbon sequestration, and circularly, protection from cyclones [46–48]. Thus, it is timely to assess the risk of tropical cyclones under climate change to ecosystems [49] and integrate this knowledge into the socio-economic risk assessment methodologies established in the context of the Intergovernmental Panel on Climate Change (IPCC) [2] and Sendai Disaster Risk Reduction [50] frameworks. We propose here using the open-source CLIMADA risk modelling platform [51] to investigate the vulnerability of global coastal ecosystems to tropical cyclones at a global scale based on the panarchy principles [52]. More precisely, we will study the spatial frequency distributions of the storms under current climatic conditions (see illustration

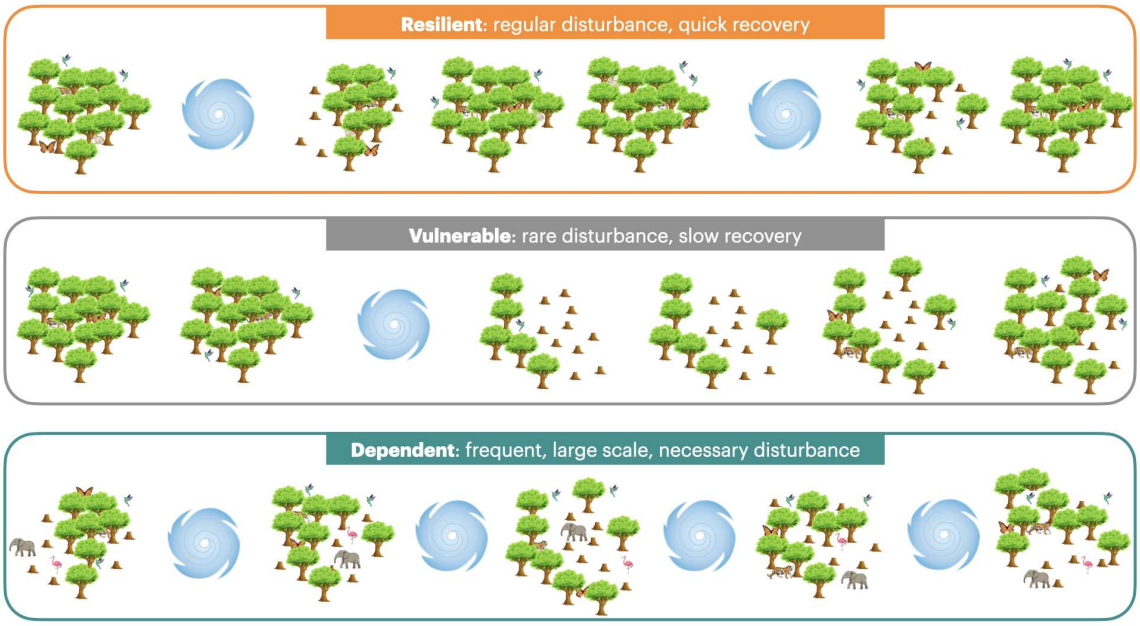

Figure 1: Illustration[1] of the vulnerability of ecosystems to extreme weather events derived from the historical occurrence patterns taking a long-term systemic perspective. Ecosystems are either resilient (able to recover quickly and remain stable), vulnerable (have difficulty to recover and might easily shift), or dependent (require regular, large scale disturbances to sustain themselves).

[1] Clipart pictures taken from https://www.freepik.com

Fig. 1) in order to then derive where climate change might put ecosystems at risk of modification. Firstly, a probabilistic tropical cyclone model [53] is used to establish a present baseline that allows for determining the average level of disturbance locally, both in terms of intensity and frequency. Secondly, depending on the historical frequency of tropical cyclones and the potential adaptation of ecosystems to these disturbances, we classify coastal areas as resilient (able to recover quickly and remain stable), vulnerable (have difficulty to recover and might easily shift), or dependent (require regular disturbances to sustain themselves). Thirdly, we use a probabilistic set of projected tropical cyclones [13] to calculate the change in frequency for each ecosystem in 2050 under the climate change pathway Representative Concentration Pathways (RCP) 8.5 (compatible with policies in 2020 [54]) and identify ecosystems at risk.

2 Results

2.1 Tropical cyclone disturbance on ecosystems

For modeling the tropical cyclone disturbance, we use a probabilistic set of tracks generated statistically using as baseline the historical records from 1980 − 2018 [53]. Furthermore, we use the ecoregions introduced by Dinerstein *et al.* [55] as the definition of global ecosystems for which tropical cyclone disturbance is assessed. As shown in Fig. 2, tropical cyclones are regionally confined phenomena and it is appropriate to use ecoregions (indicated by polygon shapes) as they combine both biological composition, climatic condition, and geographic information. In contrast, considering biomes (indicated in colors in Fig. 2)

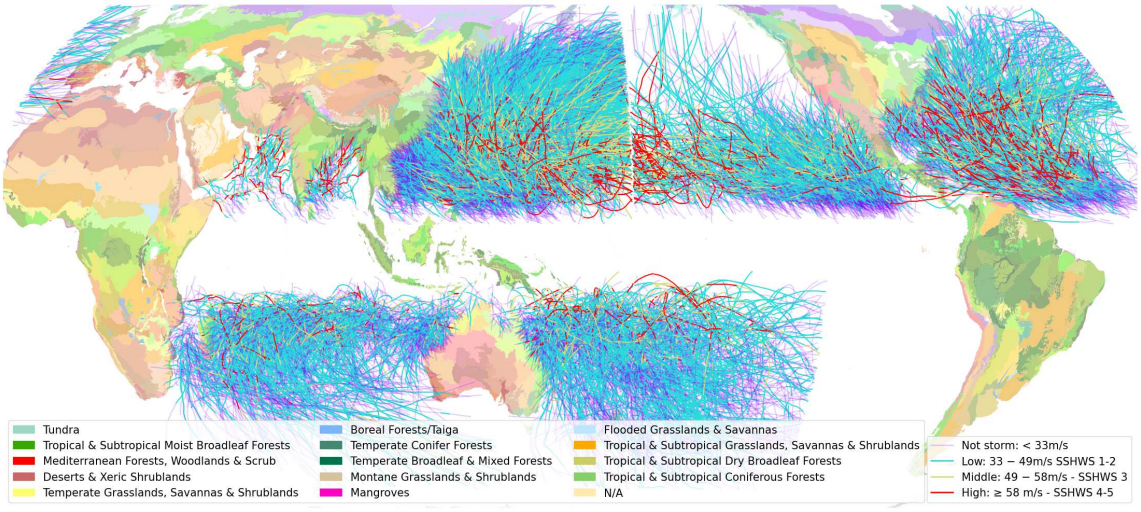

Figure 2: Subset of 10000 simulated probabilistic tropical cyclones track from the STORM dataset [53] based on the historical track records from 1980 − 2018 forming the baseline climate scenario to determine the ecosystems' (ecoregions – shown as polygons – divided into 15 biomes – indicated by colors) vulnerability. The sustained one minute wind speed intensities are divided into low(L), middle(M), and high(M) intensities based on the Saffir-Simpson scale (SSHWS) (here shown for the center of each cyclone track). Note that the sharp boundaries of certain tracks at the anti-meridian are an artifact of the STORM dataset.

would not be suitable for studying the relationship between tropical cyclones and ecosystems. For instance, Tropical & Subtropical Moist Broadleaf Forests can be both found in the Amazon and Congo basins, which are storm-free, as well as in the Mexican peninsula and the Philippines that experience some of the highest tropical cyclone activity worldwide. For each cyclone track, we derive a corresponding wind field which is used as a proxy for their intensity. Since it is expected that the impact at different intensities differs vastly, we separate wind speeds into three categories: low (L), middle (M), and high (H) intensity storms (see Fig. 2 and Section 3). For each grid point on land, and each storm category, we derive the average annual frequency as shown in Supplementary Fig. 1. We find that 12.7% of the area of all terrestrial coastal ecosystems experience a tropical cyclone of at least low intensity once every 250 years over their territory on average (c.f. Supplementary Fig. 3). This is the basis for determining the vulnerability of ecosystems to tropical cyclones.

2.2 Vulnerability

We define the vulnerability to tropical cyclones for each ecoregion for the three intensity categories (low, middle, and high) as being either vulnerable, resilient, or dependent [56] (c.f., Fig. 1) based on the panarchy principles [52] and ecosystems dynamics model assumptions [31]. A vulnerable ecosystem has rarely been exposed to the disturbance and the damages of tropical cyclones are high since it has likely not evolved resilience and recovery mechanisms. In contrast, a resilient ecosystem has historically frequently been exposed to tropical cyclone disturbance and thus has evolved to cope with the destructive effects and recovers on time scales of roughly the historical disturbance frequency. Finally, we define an ecosystem as dependent when the disturbance is naturally present at high frequency over most of its geographical

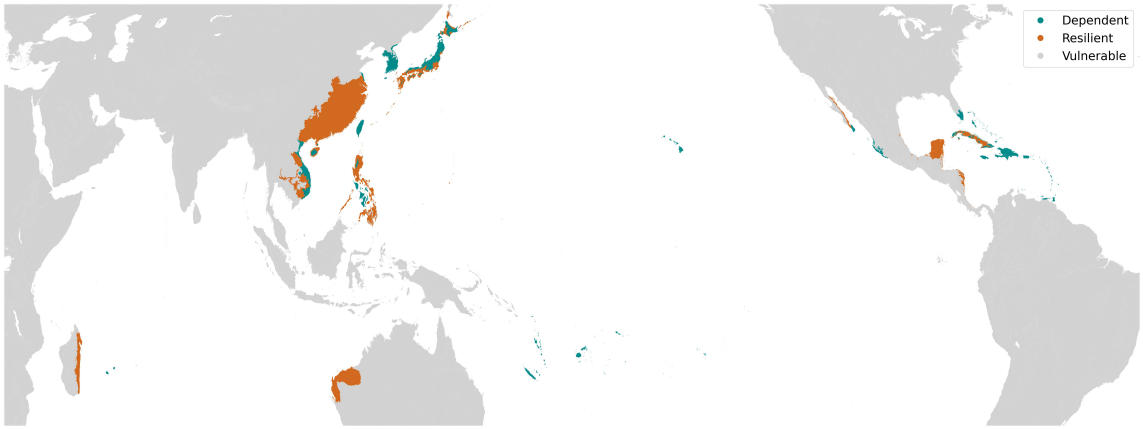

Figure 3: Vulnerability of ecosystems to tropical cyclones being resilient, dependent or vulnerable as defined in Sec. 3.2. An ecosystem is labeled as resilient if it is resilient to at least one storm intensity, and dependent if it is dependent to at least one storm intensity, but not resilient (c.f. Table 2 for the detailed parametrization values). A map for each storm intensity is available in Supplementary Fig. 4.

extent, and thus its natural ecological dynamics may depend on these regular disturbances [57]. Based on these definitions and using the tropical cyclone frequency maps (Supplementary Fig. 1), we determine the vulnerability of all ecoregions. The exact parametrization is described in the methods section 3.2.

Using this classification, we find that resilient ecoregions cover an area of 1.7 million km^2, the dependent an area of 11.86 million km^2 and the vulnerable ones affected by cyclones an area of 177.8 million km^2. Together, the resilient and dependent ecosystems represent 2.4 % of the surface of all terrestrial ecosystems and 19 % of those affected by storms at least once in 250 years. Of the 802 ecoregions worldwide – of which 269 are affected by tropical cyclones, c.f. Supplementary Fig. 3– 180 are vulnerable. Moreover, 24 ecoregions were classified as resilient and 47 as dependent, where we remark that all ecoregions resilient or dependent on a higher-intensity storm, turn out to be dependent on the corresponding lower-intensity storms. Most of the resilient and dependent ecoregions are concentrated in the larger East-Asian regions, the Caribbeans and the Central American peninsula, the Pacific islands, and Madagascar as shown in Fig. 3. Among those, we found that only 5 (Ogasawara subtropical moist forests, Luzon montane rain forests, Luzon rain forests, Mariana's tropical dry forests, and Nansei Islands subtropical evergreen forests) are fully adapted to high (H) intensity (winds larger than 58m/s) storms. For the resilient ecosystems, the average return period of tropical cyclones is 14 (L), 20(M), and 19(H) years, for dependent ecosystems 11(L), 13(M), and 17(H) years, and for vulnerable (and affected) ecosystems 88(L), 108(M) and 117(H) years. These return periods can be interpreted as average upper ends to the recovery time of the ecosystems. A detailed list of all vulnerabilities including the average return period is given in Supplementary Table 1, and the vulnerabilities per category are shown in Supplementary Fig. 4.

2.3 Projected effects of climate change

We compare the regime of tropical cyclones for the three intensity categories in the baseline $1980 - 2018$, and in 2050 under the high-emission scenario RCP8.5 (compatible with policies in 2020 [54]). For the future climate, we consider the outcome from 4 Global Circulation Models (GCMs) and use the median return period at each grid point as described in Section 3. For each ecosystem, we compare the change in the

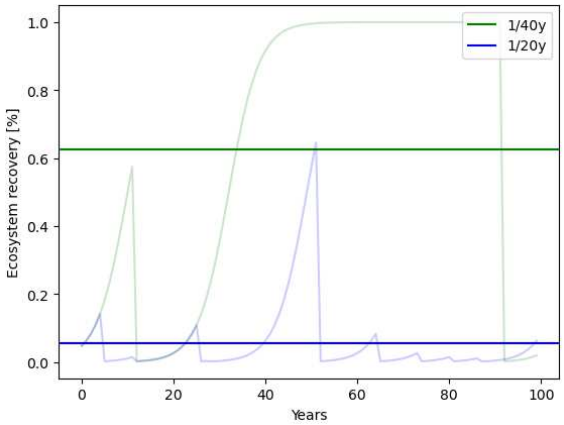

Figure 4: Illustration of the effect of a doubling of frequency on an ecosystem equilibrium state. At a larger frequency, the system cannot recover and is eventually pushed into a new state. A simple sigmoidal recovery is assumed for illustrative purposes.

average tropical cyclone frequency. Taking into account their vulnerability, we then identify ecosystems that are likely to be at risk. We assume that a change in frequency can lead to a change in ecosystem composition as the dynamical processes required for recovery are not anymore synchronized with the disturbance regime [5, 18, 25, 36, 52]. In general, if the average frequency of tropical cyclones in a given region increases significantly, the system would be in turn disturbed in a way that does not allow a return to the initial state as illustrated in Fig 4. In addition, for dependent regions, a significant decrease can also change the ecosystem's composition as the vital disturbance is reduced.

We find that 204 ecoregions see a significant increase in frequency, 5 are confronted with a significant decrease in frequency, and 28 are newly affected. While the significant decrease in frequency is confined to the Caribbeans and the larger Antilles, the increase is found all over the world (see Fig. 5): in Oceania, Madagascar and the East-African coast, the Arabian peninsula, North-West Africa and the Iberian peninsula, South-East and North-West North America, the Caribbeans, Northern South America, the Pacific islands, East Asia, and the Philippines. Overall, the combined surface of all ecosystems at risk from tropical cyclones in a changing climate amounts to 13% of the combined area of all land ecosystems worldwide. Of these, the newly affected ones contribute 3% of the total area. A detailed list of all values for all ecoregions is given in Supplementary Table 2.

On average, ecosystems at risk from tropical cyclones in a changing climate are subject to a 55%, 98%, and 84% increase in frequency for the low, middle, and high category storms, respectively. This results for instance in a reduction of the average time for recovery of resilient systems to high-intensity storms from 19 to 12 years (c.f. Table 1). Geographically speaking, for the high-intensity storms, only ecoregions around China, Japan, Korea, and the Philippines are projected to be at risk from a significant frequency increase. Newly affected and at-risk regions are found in Madagascar, some Pacific islands, and in the China, Japan, and Korea regions. For middle-intensity storms, the same regions are potentially affected. Furthermore, regions in North America and Oceania are newly affected, while the Pacific islands and the Antilles are at risk from an increase in frequency. For low-intensity storms, large areas are newly affected or subject to a significant increase in frequency. Moreover, few dependent ecoregions in the Caribbeans are confronted with a significant decrease in frequency. Detailed maps for low, middle, and high-intensity storms are shown in Supplementary Fig. 5.

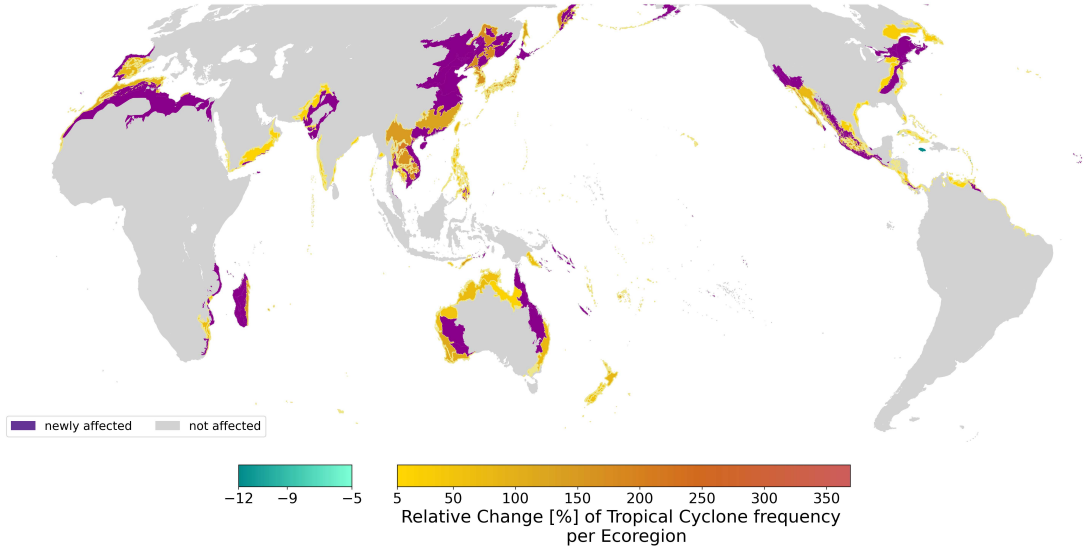

Figure 5: Map of the change in tropical cyclone frequency for all ecosystems at risk in 2050 under RCP8.5 compared to their vulnerability as defined by the tropical cyclone activity in $1980 - 2018$ (see Supplementary Fig. 4, Supplementary Fig. 1, and Table 2). Those ecoregions for which the frequency increases from below 0.004 (return period of 250 years) to above this threshold are labeled as newly affected.

Storm intensity	Year	Resilient [y]	Dependent [y]	Vulnerable [y]
Low	2020	14	11	88
	2050	**12**	**8**	**77**
Medium	2020	20	13	103
	2050	**11**	**9**	**89**
High	2020	19	17	117
	2050	**12**	**9**	**109**

Table 1: Table of the average recovery time in years per storm category for the historical baseline $1980-2018$ and the average storm return period in years in 2050 under climate projections RCP8.5. In all cases, storms are more frequent on average in 2050 than in 2020.

3 Data and Methods

3.1 Tropical cyclones

Tropical cyclones are described as a probabilistic set of wind fields of maximum 1-minute sustained winds at 10 meters above the ground. Thus, in this study, we consider the effect of tropical cyclone winds on ecosystems and do only implicitly consider the precipitations, surges, landslides, or floods that often are caused by tropical cyclones. The wind fields are computed using the Holland model [58] from a probabilistic tropical cyclone tracks (time series of the position, pressure, and maximum wind speed of the center of the cyclones) as implemented in the risk modeling platform CLIMADA [51] . The probabilistic set of tracks is obtained from the STORM model [53], a statistical model based on the observational data from the IBTrACS [59] database (c.f. Fig. 2). As the baseline we consider a set of $10,000$ years of simulated tropical cyclones for climate conditions representative for the period $1980-2018$ [53] (for a total of $144,000$ tracks worldwide). We also consider $10,000$ years of simulated tropical cyclones for climate conditions in 2050 under the emission scenario RCP8.5 [13]. In both cases, each storm in the $10,000$ year set is thus assigned an annual occurrence probability of $1/10,000$. For the future scenario, we consider the outputs from four global circulation models (i.e. four times $10,000$ years of data). All the metrics reported in the results section are then the median over all four models.

Since maximum sustained wind speed from tropical cyclones can range from 20m/s to 100m/s or more, their destructive power can vary a lot. Thus, we divide the tropical cyclone wind intensities into three categories low (L), middle (M), and high (H) based on the Saffir-Simpson scale (SSHWS):

- Low : $33-49$m/s - SSHWS 1-2
- Middle: $49-58$m/s - SSHWS 3
- High: ≥ 58 m/s - SSHWS 4-5

These correspond to the percentiles 72, 93, and 98 of the used STORM dataset wind speeds for the current climate (see Supplementary Fig. 2). For each intensity category, at each grid location (global resolution of 150arcsec on land), we then obtain the average annual occurrence probability (frequency), or equivalently its inverse, the average return period. Note that there are between $100-150$ storms per year on the planet.

The average return periods for each intensity category for the baseline climate are shown in Supplementary Fig. 1. We do not consider return periods larger than 250 years here. We assume that ecosystems, and in particular other environmental factors such as temperature, humidity, or precipitation, as well as Human factors such as urbanization, change on timescales much faster than would be required to be in dynamical equilibrium with events that occur less than with 0.4% probability, i.e. with a return period of 250 years.

3.2 Vulnerability of ecosystems

It is assumed that an ecosystem must be in a stable equilibrium with the external perturbations acting on it. To describe the vulnerability of ecosystems to tropical cyclones, we thus use a purely statistical approach based on the description of the tropical cyclone's temporal patterns. This approach is strongly inspired by the well-established framework for extreme weather risk analysis [2, 51].

We base the logic of the classification on the following arguments. An ecosystem is vulnerable to tropical cyclone winds of a given intensity if it has rarely been exposed to disturbance and thus when it is, the damages have potentially strong consequences and might strongly modify the ecosystem which has not evolved resilience and recovery mechanism. A resilient ecosystem has frequently been exposed to the disturbance and thus has evolved to recover and cope with the negative effects on time scales of at least the average disturbance frequency. A dependent ecosystem is experiencing regular tropical cyclones on almost the entirety of its territory. Or, conversely, the ecosystems exist only there where there are regular tropical cyclones passing. Hence, it is likely that tropical cyclones play a key role in maintaining the ecosystems in their current state.

Vulnerability	Area affected	Recovery time (years)	Frequency change risk
Resilient	$20-80\%$	$1-20$	$+20\%$
Dependent	$>80\%$	$1-20$	$\pm 5\%$
Vulnerable	$<20\%$	>20	$+10\%$

Table 2: Parametrization of the vulnerability and the climate change risk indicators. Resilient ecosystems have a minimum average return period of 20 years over $20-80\%$ of their area, and an increase of 20% of the return period over that area results in climate risk. Dependent ecosystems have a minimum average return period of 20 years over $80-100\%$ of their area, and an increase or decrease by 5% of the return period over that area results in climate risk. Ecosystems are vulnerable otherwise, and an increase by 10% of the return period over at least 20% of the area results in climate risk. Ecosystems with return periods longer than 250 years are always vulnerable, and newly at risk if the frequency crosses this threshold.

Based on these definitions and using the tropical cyclone frequency maps shown in Supplementary Fig. 1, we classify the ecoregions' vulnerability according to the following parametrization. A regular disturbance is assumed if it is with a frequency of a least $f = 1/20$ years. The surface area threshold of resilience is $x_r = 0.2$ of the total territory and for dependency $x_d = 0.8$. Thus, an ecosystem is *(*dependent) on tropical cyclones if at least 80% of its territory is affected on average at least once in twenty years, i.e., with a return period of 20 years. An ecosystem is *resilient* to tropical cyclones if between $20\% - 80\%$ of its territory is affected on average at least once in twenty years. An ecosystem is *vulnerable* otherwise.

Concretely, we compute the percentile distribution of the average tropical cyclone frequency per category of all the grid points inside of the polygons defining each ecoregion. Then, for each intensity category, we consider the 20th and 80th percentiles p_{20} and p_{80}. If $p_{20} > f$, then the system is dependent, if $p_{80} > f$ the system is resilient, and vulnerable otherwise.

3.3 Climate Change

For the climate change assessment, we use the probabilistic storm dataset for 2050 under RCP8.5 as described in Sec. 3.1 to compute the average frequency for low, middle, and high-intensity storms at each grid point for each of the four climate models. For each ecoregion, we then compute the median of the percentile distribution p^c of the average tropical cyclone frequency per category. To determine whether an ecosystem is at risk, we compare these percentiles p^c with the percentiles p for the historical climate depending on their vulnerability. For vulnerable and resilient systems, we compare p_{80} and p^c_{80}, while for dependent systems it is p_{20} and p^c_{20}. We then compute the relative ratio $r = (p - p^c)/p$ to obtain the relative change in tropical cyclone frequency. Ecosystems are then classified as at risk if the ratio $r > 0.1$ for vulnerable systems, $r > 0.2$ for resilient systems, and $r > 0.05; r < -0.05$ for dependent systems. We thus assume that resilient systems are able to cope better with an increase in frequency than vulnerable ones, and we assume that dependent systems are likely fine-tuned to the storms' perturbation and thus are more sensitive to changes in frequency.

The thresholds, summarized in Table 2, are a clear modeling choice that will impact the classification of ecosystems at risk. For instance, if we consider that resilient and dependent ecosystems require double and half the relative changes as the vulnerable ones, we find that the number of ecosystems at risk varies from 240 to 190 if the thresholds change from 10% to 30% as shown in Supplementary Fig. 6. Thus, there is a certain arbitrariness in our definition of ecosystems at risk, and would strongly benefit from future inputs from field studies.

4 Discussion

Contrasting with many local assessments [21, 26] associated with specific events [25], here we derived potential vulnerability relationships between the large-scale natural hazards of tropical cyclones, and ecosystems at a global scale. We used our approach inspired by the panarchy principles [52, 60] and based on the frequency patterns of the storms to characterize ecosystem vulnerability under present climate conditions. We identified ecosystems susceptible to being exposed to transformative pressure from tropical cyclone pattern changes due to a warming climate as early as 2050. Thus, we expanded the study of extreme climate risks from the extensively studied socio-economic realm [2, 61–65] to include ecosystems complementing a growing body of research on the topic [17, 49, 66]. We found that climate change might significantly affect a large number of coastal ecosystems through tropical cyclone patterns change. Most ecosystems currently exposed to tropical cyclones will be confronted with an increase in their frequency and intensity thus narrowing the maximum boundary for their recovery. This might have severe consequences for the ability of these ecosystems to prevail under future climatic conditions and is likely to result in further biodiversity losses [67] as is accounted for in climatic envelop models which do not fully account for extreme weather pattern changes [1, 68, 69]. Furthermore, this might reshape considerations for the viability of coastal ecosystem-based adaptation measures to climate change [46, 70, 71], since the ecosystems themselves are at risk of being transformed. While previous studies have done local or ecosystem-specific assessments or projections [12, 16, 30], our study offers a global assessment of ecosystem disturbances by tropical cyclones under climate change.

The change in geographical distribution of cyclones may result in several new ecosystems being exposed, in particular further away from the tropics with an increase in storms undergoing extra-tropical transitions [72–75]. Forests and other coastal ecosystems at higher latitudes are not resilient to tropical cyclones since, in contrast to low latitude areas, no evolutionary forces made them able to withstand those perturbations [12]. Among the potentially newly affected ecoregions, some are located far outside of the tropics, where recent tropical storms have already strongly impacted ecosystems [39]. For those regions historically hit by storms, the general projected increase in storm frequency and intensity might cause particularly important changes to ecosystems as illustrated by the recent events Ana in Mozambique in 2022 [76] or Amphan in Bangladesh in 2020 [77, 78]. A few ecosystems might in contrast have to deal with a significant decrease in frequency, which also could lead to a change in composition due to the disappearance of a functional disturbance. This might change the natural ecosystems' dynamics and may influence their associated biodiversity [67].

The major strengths of our approach are the global applicability, the exclusive use of available open-data sources, the possibility to extrapolate the results to future climates, the computational simplicity, the parametric flexibility, the purely statistical basis of the approach, the potential for direct generalization to other climate hazards, and the seamless integration into the established IPCC risk [2, 79] framework. While we here concentrated on the characterization of vulnerabilities and risk under future climates, the approach also provides estimates of upper bounds to recovery times from the average return periods of the storms (c.f. Supplementary Table 1). These can potentially inform dynamical ecosystem models [35], or help ecosystem management practitioners anticipate potential regime changes. Furthermore, since the proposed framework is seamlessly integrated into the open-source risk modeling platform CLIMADA [80], it is easily applicable to any natural hazard, such as floods, earthquakes, extreme heat, or wildfires. For several of these hazards, open data is readily available from the ISIMIP [81] or the CLIMADA data APIs [80]. Note however that for some hazards the assessment might be more involved since the intensity and frequency might depend on the ecosystem itself – think for instance of wildfires whose ignition probability depends on the available biomass fuel. This challenge could be tackled by qualifying ecosystems not only as resilient, vulnerable, or dependent but also as drivers of the hazard.

Our study represents one step towards a systemic, global, and long-term understanding of ecosystems' vulnerability to tropical cyclones. It has some limitations that open perspectives for more detailed quantification in the future. The proposed statistical approach based on the hazard distribution alone does

not contain any ecological function understanding as this is not yet feasible at the global scale [82]. Our approach hinges on the assumption that ecosystems have co-evolved with the extreme disturbances at their location. However, given the major anthropogenic pressure on all planetary ecosystems in the last centuries [83, 84], it is from the beginning only partially valid to assume ecosystem co-evolution with tropical cyclone disturbances. Furthermore, the fact that tropical cyclone activity has fluctuated over time inherently requires making an arbitrary choice on the relevant timescales. We focused on timescales relevant to the current rapidly changing climate. In addition, several semi-informed, partially normative, parametrizations choices had to be made, which may affect the outcomes: the definition of the storm intensity categories, the choice of spatial extent for the vulnerability definitions, the choice of return period maxima for the resilient and dependent ecosystems, as well as the thresholds for the significant changes in a changing climate. Furthermore, we only considered the changes in the annual frequency distributions. Thus, we did neither consider the potential effects on ecosystems from a change in the seasonality of tropical cyclones [85], nor do we account for the clustering in time properties, which may follow multi-year patterns in several regions [86]. We also did only implicitly consider the resulting storm surges, heavy precipitations, landslides, or floods which would be explicitly needed for a mechanistic understanding of the impacts of tropical cyclones on coastal ecosystems. We further did not consider similar strong wind hazards such as winter storms, gales, or Derechos that might co-affect certain ecosystems. Finally, our study is bound by the limitations of the STORM model itself in representing current and future tropical cyclone activity [9, 13, 87], and is bound to the specific ecoregion boundaries defined by Dinerstein *et al.* [55].

In conclusion, our study shows how changes in tropical cyclone patterns under a changing climate can impact a large number of coastal ecosystems by the year 2050. The identified potential strong impact of tropical cyclone pattern changes alone indicates the need for the more explicit inclusion of the effect of extreme weather hazards on global ecosystems in earth-system models to improve our understanding of climate projections [82, 88]. In the context of ecosystem management to mitigate and adapt to climate change, our findings show that we need to not only consider immediate damages from a tropical cyclone event but also think about multi-year shifts in ecosystem trajectories in a changing climate. This includes the need to better assess and report the consequences of storms for ecosystems at a global scale. Our study does not directly allow us to conclude whether the increased risk to ecosystems translates into a decrease or an increase in ecosystem service provisions, yet based on ecosystem disturbance studies, a strongly perturbed ecosystem is expected to be less capable of sustaining biodiversity, carbon storage or providing flood protection [89]. Considering the further pressure on coastal ecosystems from both average climate variables changes [1] and anthropogenic activities [83], it is worrying to add yet another potential threat to the list in the form of tropical cyclones.

Data Availability

For this study, the Python (3.8+) version of CLIMADA (3.2+) can be used [90]. The source code is openly and freely available under the terms of the GNU General Public License Version 3[51].

Code Availability

Code to reproduce the results of this paper will be made available upon acceptance.

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Author contributions statement

C.M.K. conceived and designed the study, analyzed the results, and wrote the manuscript. L.P. contributed to the writing, interpretation of the results, and general framing of the study. L.V. contributed an early study. D.N.B. provided input on the overall framework. All authors (L.P., D.N.B., L.V., C.M.K.) reviewed and edited the manuscript.

Competing interests

The authors declare no competing interests.

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